Exact finite-depth crossings in a Sierpinski–Hanoi free-fermion spin model

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Abstract

Chapman and Flammia introduced a three-body spin model on a finite Sierpinski sieve whose fixed symmetry sectors map to free Majorana fermions. At recursion depth five, their single-particle spectrum contains two marked levels that were degenerate to numerical precision, and their locations were conjectured to approach zero with depth. We resolve this finite-size question in the sector where every canonical cycle symmetry is $+1$. A signed three-copy Schur reduction gives a division-free recurrence for three scalar polynomials and an exact quotient for the complete characteristic polynomial. On its flat-band locus, the recurrence can be solved at arbitrary depth. For every $k \geq 3$ there is a unique nonzero pair of transverse crossings

$$J_k^{\pm} = \pm 3^{-(k-3)/2}, \qquad \lambda_k = \frac{1}{2} 3^{-(k-3)/2},$$

where λ uses the normalization of the source spectrum. Each eigenvalue sign has multiplicity $(3^{k-2}+1)/2$ at the crossing, and $|J_{k+1}/J_k| = 1/\sqrt{3}$ exactly. Thus the two depth-five landmarks occur at $J = \pm 1/3$ and $\lambda = 1/6$ and are exact crossings rather than avoided crossings. We also separate windows that follow the crossing from the fixed intervals $[1/4, 1/2]$ and its negative: on those fixed intervals the tracked minimum is zero only at $k = 5$, with a constructive positive bound otherwise. All statements concern a finite open, single-particle, all-cycle-$+1$ sector; they do not establish a many-body gap or a thermodynamic phase transition.

1 Introduction

Mappings from interacting Pauli Hamiltonians to free fermions turn an exponentially large spin problem into a single-particle spectral problem. The frustration-graph framework of Chapman and Flammia characterizes an important class of such mappings in terms of line graphs and their root graphs [1]. Related graph-based constructions extend the class of hidden free-fermion models and connect their spectral properties to graph invariants [2, 3].

The final example of Ref. [1] is a three-body XYZ model on a recursively constructed Sierpinski sieve. In a fixed cycle sector its frustration graph is a Hanoi graph and its root graph supports a free Majorana hopping problem. A local Pauli field attaches pendant Majorana modes to the root graph. At depth $k = 5$, two circled features in the numerical single-particle spectrum become degenerate within the reported precision. Their locations move toward $J = 0$ as the depth grows, suggesting a finite-size trace of the scale symmetry of the infinite fractal. The source does not determine whether the finite-size meetings are exact, nor does it give their locations or an all-depth characteristic recurrence.

Self-similar recurrences also occur in other statistical models on Sierpinski and Hanoi lattices [4], but those recurrences do not include the cycle-sector signs, local-field pendants, or Majorana observable considered here. A bounded search of the defining work and the related primary works above found no earlier expression for the signed recurrence or the crossing law derived below. This is necessarily a bounded literature statement, not an exhaustive priority claim.

Contributions. The assumptions and results are as follows.

1. We fix the finite open geometry, the canonical all-cycle-+1 sector, and the source normalization before taking a Schur complement.

2. We prove that the signed three-copy composition has a depth-independent gauge representative. Its three-corner response closes on two rational scalars, which admit a cancellation-safe polynomial lift (A_k, U_k, D_k).

3. We solve this recurrence exactly on the cycle-flat locus and obtain the complete residual factor. This proves the location, multiplicity, channel, transversality, and $1/\sqrt{3}$ scale ratio of the unique nonzero crossing.

4. We distinguish a moving continuation window from the fixed intervals used to define an avoided-crossing question, and derive a positive coefficient-norm bound whenever the exact crossing is outside the latter.

No minimality claim is made for the three-scalar state, and no no-go statement is made about every possible relation involving characteristic polynomials alone.

2 Model, sector, and normalization

Let $k \geq 1$ denote the recursion depth, with $k = 1$ a single upright three-qubit cell. For each ordered cell (i, j, ℓ), the Pauli labels are X, Y, Z at the lower-left, lower-right, and top vertices, respectively. In units where the three-body coupling is one, the finite open Hamiltonian is

$$H_k(J) = \sum_{(i,j,\ell)\in\mathcal{N}_k} X_i Y_j Z_\ell + J \sum_{q\in B_k} \sigma_q^{a_q}, \qquad J \in \mathbb{R}. \tag{1}$$

Here B_k contains the qubits shared by two cells, and a_q is the third Pauli label not used at q by those cells. This is precisely the local-field augmentation of Eqs. (72) and (77) of Ref. [1].

At $J = 0$, the frustration graph is the three-peg Hanoi graph H_3^{k-1}. The root graph has

$$|V(R_k)| = \frac{5\,3^{k-2}+3}{2} \quad (k > 1), \qquad c_k = \frac{3^{k-2}-1}{2} \quad (k > 2) \tag{2}$$

vertices and independent cycles, respectively. The local field appends one dedicated Majorana mode to every degree-two root vertex, excluding the three corners. We select the sector in which the canonical Hermitian generator of every fundamental cycle has eigenvalue +1. This selection fixes the cycle products of the oriented hopping graph; changes of spanning tree or edge orientation are thereafter only diagonal Majorana gauges.

Write the free-fermion Hamiltonian in the convention of the source as

$$\widetilde{H}_k = \mathrm{i}\gamma h_k(J)\gamma^{\mathsf{T}} = 2\mathrm{i}\sum_{a<b} h_{k,ab}(J)\gamma_a\gamma_b. \tag{3}$$

A Pauli term with coefficient $w \in \{1, J\}$ therefore produces an oriented entry $h_{ab} = \pm w/2$. It is convenient to remove this factor by defining the real antisymmetric matrix

$$\mathcal{A}_k(J) = 2h_k(J), \qquad C_k(z, J) = \det(zI - \mathcal{A}_k(J))\,. \tag{4}$$

The physical Hermitian matrix is $\mathrm{i}\mathcal{A}_k$. If x is one of its positive eigenvalues, the ordinate in the source figure is $\lambda = x/2$. This single-particle quantity is not a many-body excitation gap; a physical parity projection would be required for that interpretation.

3 Signed self-similarity and the Schur state

We first isolate the sign information that survives recursive composition.

Lemma 1 (Depth-independent signed composition)**.** *For every $k \geq 2$, the depth-$(k+1)$ augmented root graph consists of three prefix copies of the depth-k graph, joined at three pairwise-distinct interface vertices that acquire three new pendants. In the all-cycle-$+1$ sector, each copy is vertex-gauge equivalent to $\mathcal{A}_k$, and the quotient macrocycle has one fixed gauge orbit. In a representative of that orbit, the three effective interface signs are $(+,+,-)$.*

Proof. Prefixing the Hanoi address by $0, 1$, or 2 translates an upright cell but preserves its ordered X, Y, Z labels. Hence every internal cycle product of a prefix copy equals the corresponding depth-k product. Equality of all cycle products is exactly the vertex-gauge criterion for signed graphs, so each restriction is gauge equivalent to $\mathcal{A}_k$.

The cycle ranks satisfy $c_{k+1} = 3c_k + 1$, leaving one quotient cycle after the three copy cycle spaces are removed. Its cross-copy term pairs are

$$(01\cdots1, 10\cdots0) : Y/X, \quad (02\cdots2, 20\cdots0) : Z/X, \quad (12\cdots2, 21\cdots1) : Z/Y. \tag{5}$$

Within a prefix copy, the two interface ports have a unique side geodesic. Its length obeys $L_2 = 2$ and $L_{k+1} = 2L_k$, hence $L_k = 2^{k-1}$. Multiplying the translated Pauli terms along the three side paths cancels internal factors pairwise. The Hermitianizing phase is $-3L_k$ modulo four, whereas the unsigned Majorana loop contributes $3L_k$ modulo four. The cycle-orientation rule of Ref. [1] therefore fixes the oriented quotient product to $+1$ in the selected sector.

In the large-z Schur expansion of $zI - \mathcal{A}$, an off-diagonal port response starts with minus the oriented product along its unique shortest path. The product of the three effective copy couplings is consequently $(-1)^3(+1) = -1$. Six independent port gauges reduce every compatible signing to a representative with interface signs $(+,+,-)$, proving the claim. □

After eliminating all vertices except the three outer corners, permutation symmetry and a fixed boundary gauge put the corner Schur complement in the form

$$S_k = \alpha_k I_3 + \beta_k(\mathbf{1}\mathbf{1}^{\mathsf{T}} - I_3). \tag{6}$$

Let δ_k^{int} be the determinant of the eliminated interior block. After adding the two copy contributions at each interface, correcting the duplicated diagonal z, and eliminating the new pendant, define

$$q_k = 2\alpha_k - z + \frac{J^2}{z}. \tag{7}$$

In the representative of Lemma 1, the interface and coupling blocks are

$$\mathcal{C}_k = \begin{pmatrix} q_k & \beta_k & \beta_k \\ \beta_k & q_k & -\beta_k \\ \beta_k & -\beta_k & q_k \end{pmatrix}, \qquad \mathcal{B}_k = \begin{pmatrix} \beta_k & \beta_k & 0 \\ \beta_k & 0 & \beta_k \\ 0 & \beta_k & -\beta_k \end{pmatrix}. \tag{8}$$

Direct 3×3 elimination gives

$$\det \mathcal{C}_k = (q_k - 2\beta_k)(q_k + \beta_k)^2 \tag{9}$$

and, after the output gauge $\mathrm{diag}(1, -1, -1)$,

$$\alpha_{k+1} = \alpha_k - \frac{2\beta_k^2}{q_k + \beta_k}, \tag{10a}$$

$$\beta_{k+1} = \frac{\beta_k^2}{q_k + \beta_k}, \tag{10b}$$

$$\delta_{k+1}^{\mathrm{int}} = (\delta_k^{\mathrm{int}})^3 z^3 (q_k - 2\beta_k)(q_k + \beta_k)^2. \tag{10c}$$

The rational form is compact, but it can hide characteristic factors when a corner response has a pole. The following polynomial lift retains them.

Theorem 2 (Division-free characteristic recurrence)**.** *Define scalar polynomials* $A_k, U_k, D_k \in \mathbb{Z}[J, z]$ *at depth two by*

$$A_2 = z^2 + 1, \qquad U_2 = 1, \qquad D_2 = z. \tag{11}$$

For every $k \geq 2$*, set*

$$r_k = 2A_k + 1 + (J^2 - z^2)U_k, \tag{12a}$$
$$A_{k+1} = A_k r_k - 2, \tag{12b}$$
$$U_{k+1} = U_k r_k, \tag{12c}$$
$$D_{k+1} = D_k^3 U_k (r_k - 3). \tag{12d}$$

Then the complete characteristic polynomial of equation (4) is

$$\boxed{C_k(z, J) = \frac{D_k(A_k + 2)(A_k - 1)^2}{z^3 U_k}.} \tag{13}$$

The quotient is exact in $\mathbb{Z}[J, z]$*.*

Proof. Substitute

$$\alpha_k = \frac{A_k}{zU_k}, \qquad \beta_k = \frac{1}{zU_k}, \qquad \delta_k^{\text{int}} = D_k U_k^2 \tag{14}$$

into equations (7) and (10). Clearing the common denominators gives equation (12); adjoining the three corners gives equation (13). The derivation is initially an identity in $\mathbb{Q}(J, z)$, where the Schur inverses exist. Multiplication by the displayed denominators gives a polynomial identity, so it extends to their zero loci. The depth-two matrix verifies equation (11). Induction then places A_k, U_k, D_k in $\mathbb{Z}[J, z]$ and proves exact divisibility at each depth. In particular, flat factors cancelled from the rational corner response remain in D_k. □

4 Solution on the flat locus

Put

$$t = J^2, \qquad y = z^2, \qquad s = t + y. \tag{15}$$

Under $z = -\mathrm{i}x$, the locus $s = 0$ is the physical flat band $x = |J|$.

Lemma 3 (Restricted recurrence)**.** *For every* $k \geq 2$*, restriction to* $s = 0$ *gives*

$$A_k = 1 - 3^{k-2}t, \qquad U_k = 3^{k-2}, \qquad r_k = 3. \tag{16}$$

Moreover $r_k - 3 = sR_k(s, t)$ *exactly, with*

$$R_k(0, t) = 5^{k-2}. \tag{17}$$

Proof. The base case follows from equation (11). On $s = 0$ one has $z^2 = -t$ and $J^2 - z^2 = 2t$. If the first two identities in equation (16) hold, their t contributions cancel in r_k, leaving $r_k = 3$; the A and U updates then prove the next depth.

For the transverse quotient, differentiate at fixed t with respect to s and restrict to $s = 0$. A dot denotes this operation and $T_k = R_k(0, t)$. Since $J^2 - z^2 = 2t - s$,

$$T_k = 2\dot{A}_k - U_k + 2t\dot{U}_k. \tag{18}$$

Differentiating the two updates gives

$$\dot{A}_{k+1} = 3\dot{A}_k + A_k T_k, \qquad \dot{U}_{k+1} = 3\dot{U}_k + U_k T_k. \tag{19}$$

On the locus, $A_k + tU_k = 1$, so substitution into equation (18) yields $T_{k+1} = 5T_k$. Since $T_2 = 1$, equation (17) follows. □

The factor retained by D_k can now be counted without a genericity assumption.

Proposition 4 (Complete residual factor)**.** *Let $c_k = (3^{k-2}-1)/2$. Then s^{c_k} divides D_k exactly. If $\overline{D}_k = D_k/s^{c_k}$, then on $s = 0$*

$$\overline{D}_k = b_k z^{3^{k-2}}, \qquad b_k = 15^{(3^{k-2}-2k+3)/4} > 0. \tag{20}$$

Consequently, for $t > 0$,

$$\left.\frac{C_k}{s^{c_k}}\right|_{s=0} = (-1)^{c_k-1} b_k 3^{k-1} t^{c_k+1}\left(1 - 3^{k-3}t\right). \tag{21}$$

Proof. The D update in equation (12) and Lemma 3 show that the flat exponent obeys $c_{k+1} = 3c_k + 1$ with $c_2 = 0$, giving the stated c_k. The quotient is nonzero because $R_k(0,t) = 5^{k-2}$. The remaining coefficient obeys $b_2 = 1$ and $b_{k+1} = b_k^3 15^{k-2}$, whose solution is equation (20). Finally insert equations (16) and (20) into equation (13). All factors are nonzero for $t > 0$ except the one shown explicitly, proving both completeness and equation (21). □

5 Exact crossings and scale law

Theorem 5 (Finite-depth crossing sequence)**.** *In the finite open all-cycle-+1 single-particle sector, every integer $k \geq 3$ has exactly one nonzero positive collision on the cycle-flat locus, together with its $J \mapsto -J$ gauge twin:*

$$\boxed{J_k^{\pm} = \pm 3^{-(k-3)/2}, \quad x_k = 3^{-(k-3)/2}, \quad \lambda_k = \frac{1}{2}3^{-(k-3)/2}.} \tag{22}$$

For either physical eigenvalue sign $\pm x_k$, the generic flat multiplicity c_k increases to $c_k + 1 = (3^{k-2}+1)/2$. The additional branch is the boundary-symmetric channel. The crossing is transverse at every finite depth and

$$\frac{|J_{k+1}|}{|J_k|} = \frac{\lambda_{k+1}}{\lambda_k} = \frac{1}{\sqrt{3}}. \tag{23}$$

Proof. By Proposition 4, a nonzero collision requires and is implied by $1 - 3^{k-3}t = 0$. This has the unique positive solution $t = 3^{-(k-3)}$, proving the locations and equation (23). The exact pendant gauge $\mathcal{A}_k(-J) = G\mathcal{A}_k(J)G$ for a diagonal sign matrix G supplies the negative twin.

The factor s^{c_k} gives c_k flat modes for each eigenvalue sign. The remaining factor in equation (21) has a simple root, so the multiplicity becomes $c_k + 1$. Since $\mathrm{i}\mathcal{A}_k$ is Hermitian, algebraic and geometric multiplicities coincide.

To identify the extra branch, the boundary-symmetric eigenvalue of equation (6) satisfies the recurrence identity

$$\alpha_k + 2\beta_k = \frac{A_k + 2}{zU_k} = \frac{A_{k-1}}{zU_{k-1}}. \tag{24}$$

Its flat-locus zero is exactly the last factor of equation (21). The two orthogonal corner channels carry $A_k - 1$, which does not vanish there; the persistent cycle-flat modes have zero corner response.

For transversality, set $n = k - 3$. Differentiating the recurrence at the collision gives

$$\partial_y A_{k-1} = \frac{5^{n+1} + 6\,3^n + 1}{12} > 0. \tag{25}$$

Implicit differentiation of $A_{k-1}(J^2, -x^2) = 0$ then yields

$$\left.\frac{dx_{\mathrm{disp}}}{dJ}\right|_{J_k^+} = \frac{5^{n+1} - 6\,3^n + 1}{5^{n+1} + 6\,3^n + 1}. \tag{26}$$

The flat branch has slope one, and their leading separation in x is

$$\frac{12\,3^n}{5^{n+1}+6\,3^n+1}|J-J_k^+|, \tag{27}$$

which is strictly positive for every finite n. □

The first depths are summarized in table 1. The depth-five entry establishes that the marked source features occur at $J=\pm 1/3$ and $\lambda=1/6$, with multiplicity fourteen per eigenvalue sign. The exact sequence and its transverse coefficients are shown in figure 1.

Table 1: First members of the exact crossing sequence. Multiplicity is for one eigenvalue sign. The last column is the dispersive slope in equation (26); the flat-band slope is one.

k	$\lvert J_k\rvert$	λ_k	multiplicity	dx_{disp}/dJ
3	1	$1/2$	2	0
4	$1/\sqrt{3}$	$1/(2\sqrt{3})$	5	2/11
5	$1/3$	$1/6$	14	2/5
6	$1/(3\sqrt{3})$	$1/(6\sqrt{3})$	41	116/197
7	$1/9$	$1/18$	122	220/301

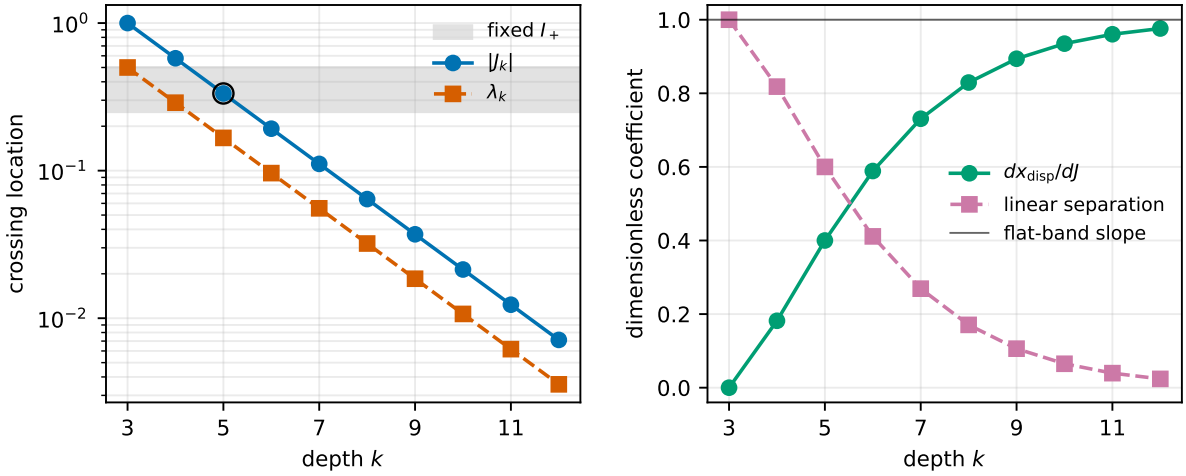


Figure 1: Closed-form finite-depth predictions, with no fitted parameters. Left: the crossing locations $|J_k|$ and source ordinates λ_k; the gray strip is $I_+=[1/4,1/2]$, and the open circle marks the depth-five source landmark. Right: the dispersive slope and the coefficient of the linear separation in x. The two branches remain transverse at every finite depth, although their slope difference tends to zero.

6 Fixed intervals versus moving windows

An all-depth crossing sequence does not imply zero minimum gap on a fixed parameter interval after the crossing leaves it. For the positive side, define $I_+=[1/4,1/2]$ and let Δ_k^+ be the minimum source-λ separation between the tracked flat and boundary-symmetric branches on I_+. Let $I_-=-I_+$ and define Δ_k^- by the gauge-related negative branches.

Set

$$F_k(x,J)=A_{k-1}(z=-\mathrm{i}x,J)=\sum_{p,q}c_{pq}J^p x^q, \tag{28}$$

and the explicit coefficient norm

$$M_k = \sum_{\substack{p\geq 0\\ q\geq 1}} q|c_{pq}|\, 2^p 4^{q-1}. \tag{29}$$

Every relevant eigenvalue obeys $|x| \leq 4$ for $|J| \leq 2$ by the row-sum bound, and therefore $|\partial_x F_k| \leq M_k$ in this box. On the flat branch,

$$F_k(J, J) = 1 - 3^{k-3}J^2. \tag{30}$$

Its minimum absolute value on I_+ is

$$d_k = \begin{cases} 3/4, & k = 3, \\ 1/4, & k = 4, \\ 0, & k = 5, \\ 3^{k-3}/16 - 1, & k \geq 6. \end{cases} \tag{31}$$

For $k \neq 5$, the mean-value theorem between $x = J$ and any zero of the tracked dispersive factor gives

$$\boxed{\Delta_k^{\pm} \geq \frac{d_k}{2M_k} > 0.} \tag{32}$$

The factor $1/2$ converts x to the source ordinate λ. At $k = 5$, $J_5 = 1/3 \in I_+$ and theorem 5 gives $\Delta_5^{\pm} = 0$. Hence the fixed-interval minimum is zero if and only if $k = 5$.

By contrast, the moving windows

$$W_k^+ = [J_k/2, 3J_k/2], \qquad W_k^- = -W_k^+ \tag{33}$$

contain the exact crossing for every $k \geq 3$, so their minima are always zero. This distinction prevents a moving finite-size landmark from being misreported as an avoided crossing on a fixed experimental control range.

7 Exact and independent checks

The model input was reconstructed in two representations: directly from the oriented Pauli cells and independently from legal Hanoi moves followed by a Krausz-root construction. At $k = 5$ both representations give 81 root edges, 69 root vertices, 39 pendant modes, and a 108-dimensional antisymmetric matrix. Every cycle product is $+1$ in the selected convention. Across the full plotted range $J \in [-2, 2]$, all 18 raster-resolvable branches of the source figure agree with the reconstructed spectrum within the resolution of the raster.

An exact $\mathbb{Z}[J, x]$ factorization at $k = 5$ contains

$$x^{24}(J^2 - x^2)^{13}(J^2 - x^2 + 3)^3(J^2 - x^2 + 5)^4 G_8(x, J), \tag{34}$$

times factors that are nonzero at the target. The relevant degree-eight factor is

$$\begin{aligned} G_8 = {} & x^8 - 9x^6 + 24x^4 - 17x^2 + 1 \\ & + J^2(-3x^6 + 19x^4 - 30x^2 + 8) \\ & + J^4(3x^4 - 11x^2 + 6) + J^6(1 - x^2). \end{aligned} \tag{35}$$

On $x = |J|$, $G_8 = 1 - 9J^2$. At $J = x = 1/3$ one has $\partial_J G_8 = 4$ and $\partial_x G_8 = -10$, recovering the slope $2/5$ and showing directly that the depth-five meeting is transverse.

The recurrence was then frozen before direct higher-depth comparisons. At $k = 6$, a separately constructed 324-mode matrix has 40 cycle-flat modes per eigenvalue sign. At $J = 1/(3\sqrt{3})$, exact arithmetic in two finite fields and both square-root embeddings gives flat-factor multiplicity 41 and rank 242 for $\mathcal{A}_6^2 + J^2 I$. Real Hermitian diagonalization finds 41 positive target eigenvalues, a separation exceeding 0.0275 from the nearest unrelated level, and the predicted order exchange. A further independent $k = 7$ reconstruction gives 122 positive target modes at $J = x = 1/9$, with the closest unrelated level separated by 0.00988 and the centered slope agreeing with 220/301. These finite calculations are adversarial checks; the arbitrary- depth statement rests on the signed-copy and polynomial inductions above.

The accompanying reproducibility package contains the exact symbolic producers, independently structured checkers, serialized graph inputs, and the short script generating figure 1. The accepted computations used Python 3.12.14, NumPy 2.3.5, SciPy 1.18.1, SymPy 1.14.0, and python-flint 0.9.0. Floating-point results are not used as certificates of the exact identities.

8 Physical interpretation and limitations

The recurrence reduces the characteristic calculation for an exponentially growing signed graph to three scalar polynomial updates. More importantly, it retains the cycle-flat factors that disappear from a naive rational corner response. The source's two numerical landmarks are therefore an exact finite-depth, scale-covariant sequence: their locations shrink geometrically, while each finite-depth meeting remains a transverse Hermitian crossing. The boundary-symmetric channel identity in equation (24) specifies which dispersive state joins the growing cycle-flat space.

The result is deliberately narrower than a statement about the full spin spectrum. It applies only to the finite open geometry and the canonical all-cycle-+1 sector. Other cycle sectors have different signed holonomies. No physical fermion-parity projection has been imposed, so λ_k is not asserted to be a many-body gap. The limit $J_k \to 0$ and the exact ratio $1/\sqrt{3}$ show finite-size scale covariance of this collision sequence, but do not prove a phase transition, critical field theory, experimental realization, or universal behavior of other fractal models. Finally, the three-scalar recurrence is constructive; its minimality among all possible recurrences has not been established.

Data and code availability

The LaTeX source, exact-certificate scripts, checker scripts, serialized graph inputs, and figure-generation code accompany this draft. A stable public archive and license statement must be selected by the author before submission; no external repository or journal submission is claimed here.

Author contributions and use of computational tools

Xiaojian Zeng is the prospective author of this draft; the final contribution statement and responsibility for the work remain subject to his confirmation. A large language model was used extensively in problem development, literature screening, derivations, symbolic and numerical calculations, code and figure production, independent computational checks, and manuscript drafting. The prospective author has not yet certified those outputs. Before submission, he must independently review the primary sources, proofs, code, numerical evidence, bibliography, disclosure, and institutional details and must make any corrections needed to assume full responsibility for the work.

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